Class 9 Physics Final Examination

Time: 3 hours **Maximum Marks:** 150

Instructions

- All questions are compulsory.
- No calculators or electronic devices are allowed.

Section A (50 marks)

(i) MCQs — 20 × 1 = 20 marks

1. A Vernier calipers has 10 Vernier divisions equal to 9 main scale divisions. If 1 MSD = 1 mm, the least count is:
   A) 1 mm B) 0.1 mm C) 0.01 mm D) 0.9 mm

2. A student notes the length as 5.20 cm while his eye is slightly left of the mark. This error is:
   A) Zero error B) Instrumental error C) Parallax error D) Personal error

3. A body moves with constant speed in a circular path. Its acceleration is:
   A) Zero B) Constant C) Changing direction D) Infinite

4. The slope of a distance–time graph of a straight line represents:
   A) Acceleration B) Velocity C) Speed D) Momentum

5. Two balls of same size but different masses are pushed with same force. The lighter ball moves faster because:
   A) It has less inertia B) It has more momentum C) It experiences less friction D) It has more acceleration due to gravity

6. A diver goes deeper into water. The pressure increases because:
   A) Water becomes heavier B) Density increases C) Height of water column above increases D) Area increases

7. An iron nail sinks in water but a large iron ship floats because:
   A) Ship is hollow and displaces more water B) Ship is lighter than nail C) Water pushes ship upward D) Iron in ship is different

8. A body displaces 1 kg of water. The upthrust on it is approximately:
   A) 1 N B) 9.8 N C) 100 N D) 0.98 N

9. In a room, hot air rises and escapes through ventilators. This is due to:
   A) Conduction B) Radiation C) Convection D) Reflection

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10. Land breeze occurs at night because:
A) Sea cools faster B) Land cools faster C) Sea becomes hotter D) Wind direction changes randomly

11. A person moves away from a plane mirror by 2 m. The distance between the person and his image increases by:
A) 2 m B) 4 m C) 1 m D) 8 m

12. A mirror has upper half concave and lower half convex. An object is placed between pole and focus of the concave part. The images formed are:
A) Both real B) Both virtual and same size C) Concave gives virtual magnified; convex gives virtual diminished D) Concave gives real; convex gives virtual

13. The focus of a convex mirror lies:
A) In front of mirror B) Behind the mirror C) At pole D) At infinity

14. Rear view mirrors are convex because they:
A) Form large images B) Show wider field of view with erect images C) Form real images D) Reflect more light

15. In a sound wave, if frequency increases, wavelength:
A) Increases B) Decreases C) Remains same D) Becomes zero

16. Sound cannot travel through vacuum because:
A) There is no air B) There is no medium for vibrations C) There is no gravity D) Temperature is low

17. According to Ohm's law, if voltage is doubled and resistance remains same, current becomes:
A) Half B) Double C) Same D) Zero

18. A wire becomes hot when current flows. This heat depends on:
A) Only resistance B) Only current C) Resistance, current and time D) Only voltage

19. Magnetic field lines never intersect because at a point they:
A) Would show two directions B) Repel each other C) Become weak D) Disappear

20. A magnet is heated strongly and then dropped on the floor. It loses magnetism because:
A) Poles are broken B) Domains get disturbed C) Weight increases D) Shape changes

(ii) Short Answer Type — 10 × 1 = 10 marks

1. A student measures the length of a table with a ruler and gets different readings each time. The ruler has no zero error. What is the most likely mistake?

2. A car moves with constant speed on a circular track. Which physical quantity is changing continuously?

3. Two bodies of different masses are dropped from the same height. Why do they reach the ground at the same time (ignoring air resistance)?
4. Why does a sharp-edged dam wall face the water while the broader side is at the bottom?
5. A steel ship floats, but a small steel ball sinks. What single physical concept explains this?
6. On a cold day, metal chairs feel colder than plastic chairs kept in the same room. Why?
7. A person standing very close to a plane mirror cannot see his full image. Why?
8. If the frequency of a sound increases but speed remains same, what must happen to its wavelength?
9. A bulb connected to a battery with thick wires glows brighter than with thin wires. Why?
10. A magnet kept near a heater for many days becomes weak. What happens inside the magnet?

(iii) Numericals — 20 × 1 = 20 marks

1. A bike moving at 12 m/s accelerates uniformly at 3 m/s^2 for 2 s. Find its final velocity.
2. A runner starts from rest and covers 18 m in 3 s with uniform acceleration. Find the acceleration.
3. A car moving at 15 m/s comes to rest after travelling 45 m. Find its deceleration.
4. A bus accelerates from 5 m/s to 25 m/s in 10 s. Find the distance travelled.
5. A stone is thrown vertically upward with velocity 15 m/s. Find the time taken to reach the highest point. (g = 10)
6. A coconut falls freely from a tree and reaches the ground in 4 s. Find the height of the tree. (g = 10)
7. A ball dropped from a bridge hits the water with velocity 40 m/s. Find the height of the bridge. (g = 10)
8. A fruit falls from a height of 80 m. Find the time taken to reach the ground. (g = 10)
9. A suitcase of weight 100 N is kept on the floor. If the area of contact is 0.5 m^2, find the pressure.
10. Find the pressure at 10 m depth in water. ($\rho = 1000$, $g = 10$)

11. A metal block displaces 4 kg of water when fully immersed. Find the upthrust acting on it.
12. A body loses 8 N of its weight when immersed in water. What is the upthrust?
13. Find the heat needed to raise the temperature of 0.5 kg water by 20°C. ($c = 4200$)
14. An aluminium utensil of mass 1 kg is heated by 30°C. Find the heat absorbed. ($c = 900$)
15. A sound wave has wavelength 0.5 m and speed 340 m/s. Find its frequency.
16. Thunder is heard 3 s after lightning is seen. Find how far the storm is. ($v = 340$ m/s)
17. A person stands 4 m away from a plane mirror. How far is his image from the mirror?
18. A charge of 24 C flows through a wire in 6 s. Find the current.
19. A bulb has resistance 3 Ω and current flowing is 2 A. Find the voltage across it.
20. A resistor of 12 Ω is connected to a 24 V battery. Find the current flowing.

Section B (100 marks)

Q1.

a) Define least count and zero error. Explain how zero error is corrected in Vernier calipers. (4)

b) A student measures the time taken by a body to fall from a height using a stopwatch. List four possible sources of error in this experiment. (4)

c) Why is precise measurement important while calculating speed? (2)

Q2.

a) Derive the relation $s = ut + \frac{1}{2}at^2$. (4)

b) Define inertia. Explain with examples the three types of inertia. (4)

c) Why do passengers fall forward when a moving bus stops suddenly? (2)

Q3.

a) State Newton's Second Law of Motion and derive the expression $F = ma$. (5)

b) Define momentum. Why is it easier to stop a tennis ball than a cricket ball moving with the same velocity? (3)

c) State the SI unit of force and momentum. (2)

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Q4.

a) Derive the expression $P = h\rho g$ for pressure in liquids. (4)

b) State Archimedes' Principle. (2)

c) Explain why a block of wood floats but an iron nail sinks in water. (2)

d) Why are dams made thicker at the bottom? (2)

Q5.

a) Describe an experiment to show that atmospheric pressure exists. (diagram) (4)

b) Define upthrust. On what factors does it depend? (3)

c) A body displaces 800 g of water. Calculate the upthrust acting on it. (3)

Q6.

a) Define specific heat capacity. Write its unit. (2)

b) Derive $Q = mc\Delta T$. (3)

c) Explain sea breeze and land breeze using convection currents. (3)

d) Why are ventilators provided near the ceiling of rooms? (2)

Q7.

a) State the laws of reflection. (2)

b) Draw a ray diagram to show image formation by a plane mirror. (3)

c) List four characteristics of the image formed. (2)

d) A person sees his full image in a plane mirror. Prove that the minimum height of the mirror required is half of the person's height. (3)

Q8.

a) What are longitudinal waves? Explain compressions and rarefactions with diagram. (4)

b) Define wavelength, frequency, amplitude and time period. (3)

c) A sound wave travels with speed 340 m/s and has frequency 170 Hz. Find its wavelength. (3)

Q9.

a) Define electric current, potential difference and resistance. (3)

b) State Ohm's Law and draw V–I graph. (3)

c) Why does a wire get heated when current flows through it? (link with heat energy) (2)

d) Calculate the current flowing through a 5 Ω resistor when 10 V is applied. (2)

Q10.

a) Draw magnetic field lines around a bar magnet. (3)

b) State four properties of magnetic field lines. (4)

c) What is magnetic induction? (1)

d) Why are magnets stored with soft iron keepers? (2)

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